

SIRD – Sports Injury Risk Detection from Video

AI-Powered Sports Movement Analysis
Infosys Springboard Internship Project

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Introduction

What is SIRD?

SIRD is an AI-enabled system that analyzes athlete movement from video to detect injury-risk indicators and present role-specific insights.

Why prevention matters

Early identification of risky movement patterns reduces downtime, improves performance continuity and supports evidence-based training and rehab decisions.

Why video-based analysis

Video is ubiquitous, non-invasive, and captures context-rich movement — enabling retrospective review, longitudinal tracking and scalable monitoring.

Problem Statement

Manual assessment limits

Manual video review is time-consuming and inconsistent across evaluators.

Hidden abnormalities

Subtle deviations in kinematics are easy to miss without quantitative measures.

Need for structured monitoring

Athletes require repeatable tracking of movement and recovery progress over time.

Fragmented information

Coaches and support staff lack a centralized, role-based view of analysis and recommendations.

Objectives



Analyze videos

Ingest and validate athlete video uploads for structured processing.



Extract movement features

Derive pose and kinematic features useful for risk assessment.



Detect risk indicators

Flag movement patterns correlated with elevated injury risk.



Maintain profiles

Track athlete history, videos and longitudinal metrics.



Role-based insights

Deliver tailored dashboards for Admin, Athlete, Coach, Physio, Scientist.



Support decisions

Provide actionable guidance for training and recovery planning.

Existing System – Limitations

- Observation-heavy workflows with human variability
- Poor historical tracking and fragmented records
- No centralized repository for multi-session comparisons
- Limited automation for extracting quantifiable movement metrics
- Hard to scale across teams and organizations



SIRD addresses these gaps by introducing automated, consistent video processing and unified role-based reporting.

Proposed System – SIRD Overview

Core user flows and capabilities

- User authentication (JWT)
- Athlete profile & organization management
- Video upload and validation
- AI-driven video processing and risk prediction
- Role-based dashboards and training/recovery guidance

01

1. Upload

Secure video submission by athlete or staff

02

2. Process

Frame-level pose extraction and feature derivation

03

3. Predict

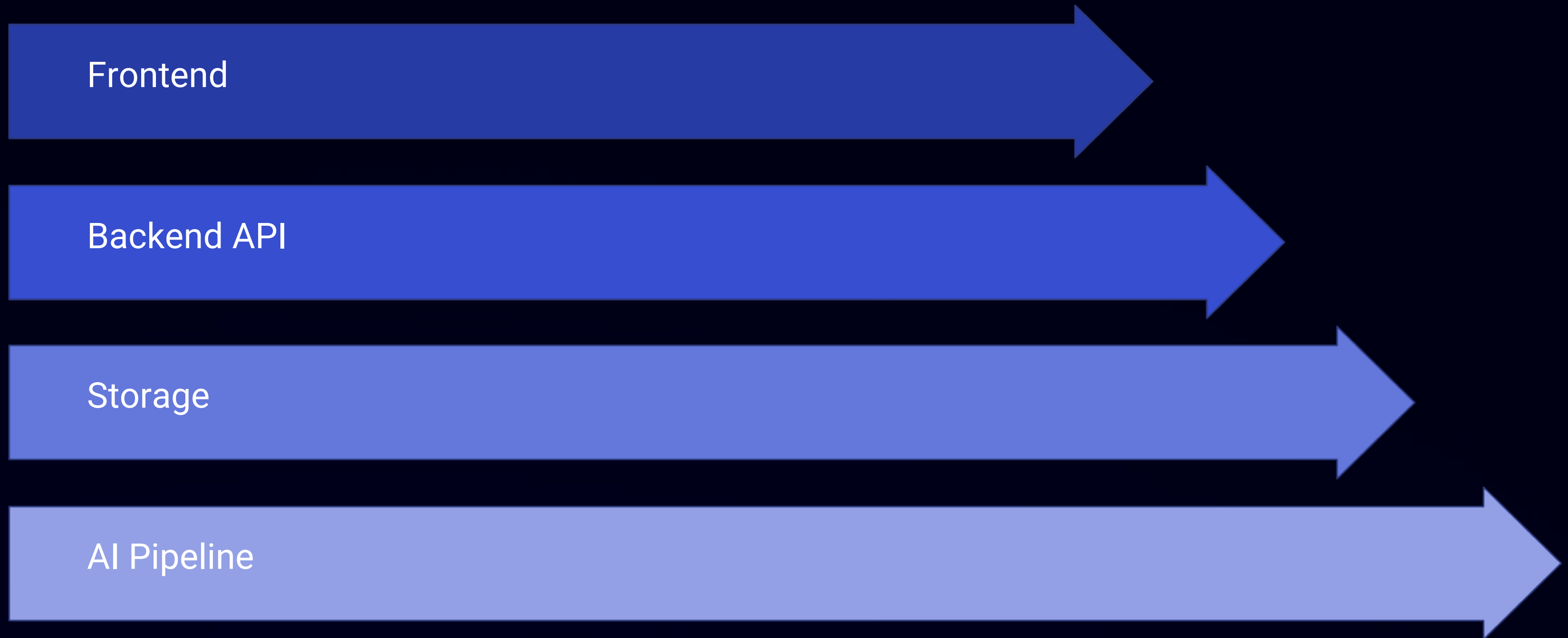
Compute injury-risk indicators and flag concerns

04

4. Report

Deliver role-specific visualizations and guidance

System Architecture



This architecture separates UI, API, persistent storage and the video/AI pipeline to enable scalability and clear responsibility boundaries.

Technology Stack



Frontend

React + Vite — responsive SPA for dashboards and uploads



Storage

PostgreSQL managed schema via SQLAlchemy



Backend & AI

Python + FastAPI for APIs and the CV/AI pipeline



Auth & Dev

JWT authentication; Git/GitHub for source control; VS Code for development

User Roles & Access



Admin

Manage organizations, generate join codes, user and system monitoring



Athlete

Upload videos, view personal progress, follow guidance



Coach

Review athlete performance, quality scores and prescribe training



Physiotherapist

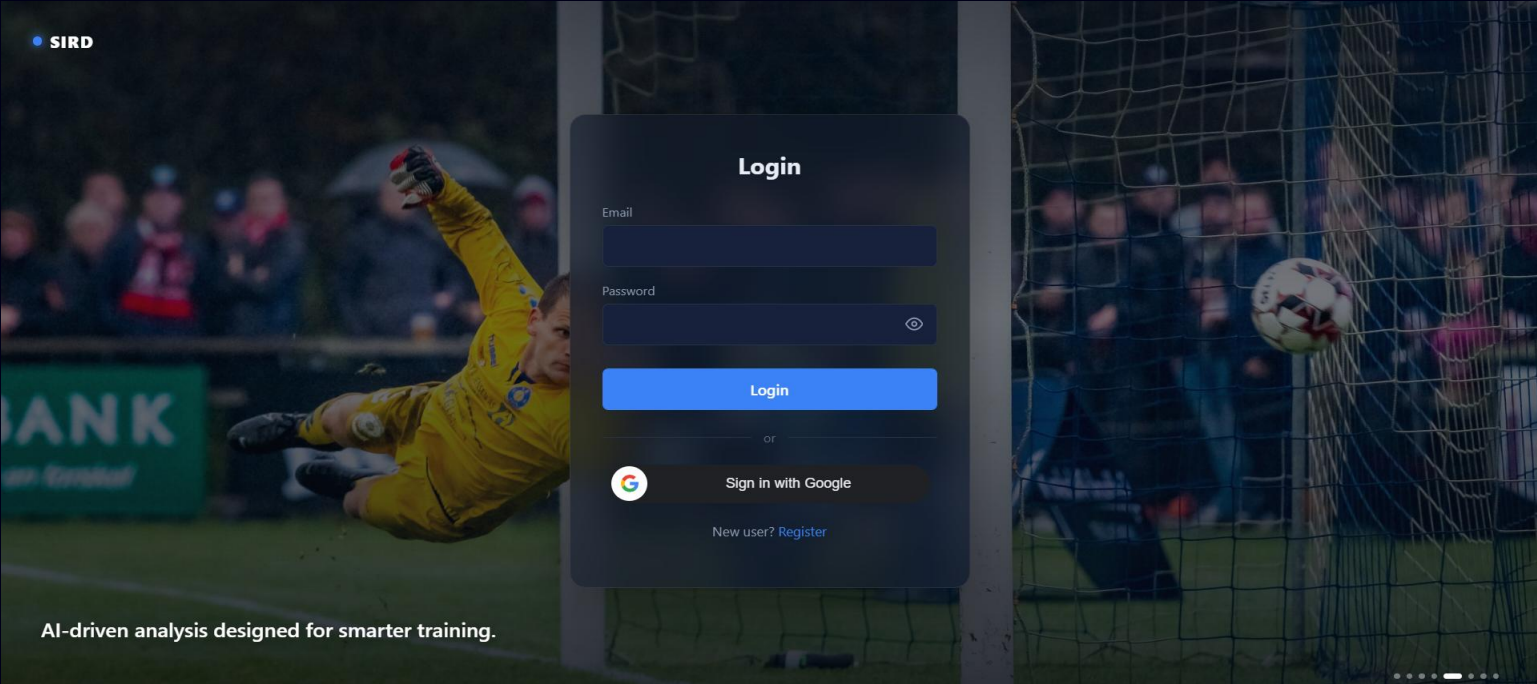
Access injury-relevant metrics, recovery plans and session notes



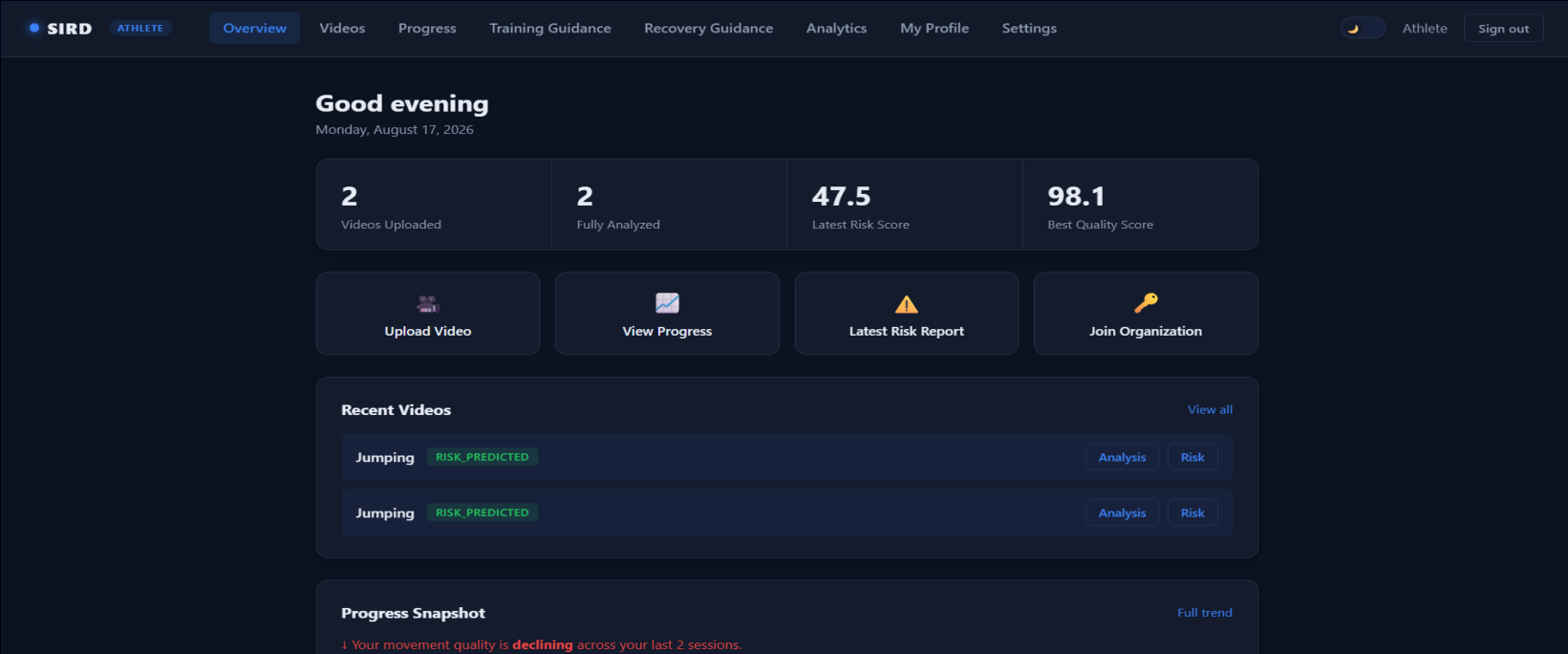
Sports Scientist

Access research-grade analysis, exportable feature data and cohort comparisons

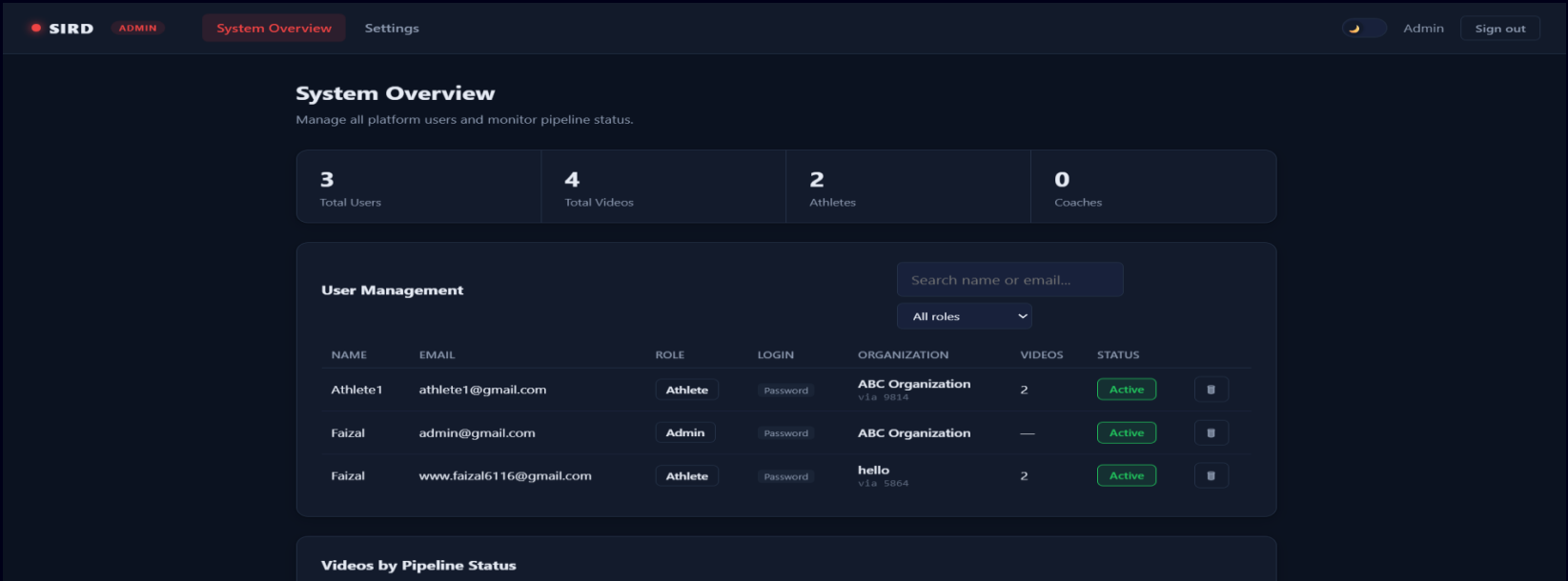
Application Screenshots



Login Page



Athlete Dashboard



Admin Dashboard

Role-Based Dashboards

SIRD

COACH

Team Overview

Settings

Coach

Sign out

Team Risk Overview

Invite athletes by email and track their injury risk status.

2

Athletes on Team

0

High/Critical Risk

0

Pending Invites

Moderate

Avg Training Load

Invite an Athlete

athlete@email.com

Send Invite

Sent Invites

EMAIL	STATUS
athlete1@gmail.com	accepted
www.faisal6116@gmail.com	accepted

Coach Dashboard

SIRD

PHYSIOTHERAPIST

Athlete Lookup

Settings

Physiotherapist

Sign out

Rehabilitation Tracking

Look up an athlete, review their movement data, and assign a recovery plan.

1

View Athlete

Faizal

Athlete

2

Videos Uploaded

Moderate

Training Load

Suggested Exercises

Based on their latest movement analysis recommendations — select exercises to add to a new recovery plan.

Wrist flexor/extensor stretch

3 sets x 30 sec hold — Forearm mobility

From: "Elbow asymmetry detected (left side lower range) — Unilateral upper-body mobility work to correct elbow movement imbalance"

Eccentric wrist curls

3 sets x 12 — Elbow tendon loading

From: "Elbow asymmetry detected (left side lower range) — Unilateral upper-body mobility work to correct elbow movement imbalance"

Physiotherapist Dashboard

SIRD

SPORTS SCIENTIST

Research Overview

Settings

Sports Scientist

Sign out

Athlete Health Research

Aggregate health index, biomechanical patterns, and injury trends across all athletes.

Export PDF

Export Excel

2

Total Athletes

4

Risk Assessments

0

Athletes At Risk

4

Videos Analyzed

Overall Athlete Health Index

Combines average movement quality and injury risk into one score. Sorted worst-first.

ATHLETE	SPORT	AVG QUALITY	AVG RISK	HEALTH SCORE	STATUS
Faizal	running	92.2	47.7	76.2	GOOD
Athlete1	jumping	94.9	46.8	78.2	GOOD

Biomechanical Deviation Frequency

How often each joint's angle fell outside the expected reference range, across all analyzed videos.

Right Knee

4

(100%)

Sports Scientist Dashboard

Future Enhancements

- **Real-Time Video Analysis**
Enable live movement and injury-risk analysis during training sessions.
- **Advanced AI Models**
Improve movement analysis and risk prediction using more advanced AI techniques and larger datasets.
- **Mobile Application**
Provide SIRD features through Android and iOS applications.
- **Cloud Deployment**
Deploy the system on cloud infrastructure for scalable access and storage.
- **Personalized Training Plans**
Generate customized training and recovery recommendations based on athlete data.
- **Long-Term Athlete Monitoring**
Track performance and injury-risk trends over multiple training sessions.

Conclusion

SIRD provides an integrated platform for video-based sports movement analysis and injury-risk assessment.

The system enables athletes to upload and monitor their performance while providing role-based access for coaches, physiotherapists, sports scientists, and administrators.

By combining video processing, AI-based analysis, athlete profiles, risk indicators, and centralized dashboards, SIRD aims to support **early risk identification, informed decision-making, and safer sports training**

Q&A

THANK YOU